LEC. 3.6: PLASMA MEMBRANE SPECIALIZATIONS

Learning goals:

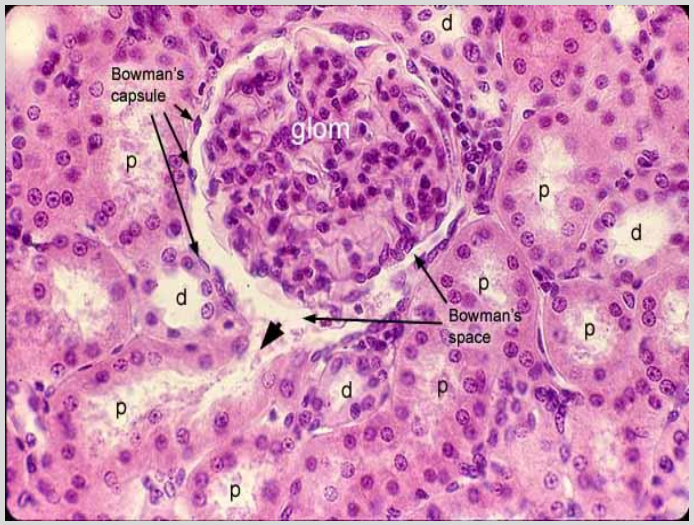
- To know about the types of PM specializations
- To understand the structure of PM specializations
- To learn the function of PM in the context of human tissues/organs

Dr. Silve Vicent (silvevicent@unav.es)

Associate Lecturer, Department of Pathology, Anatomy and Physiology, School of Medicine, UNAV
Group Leader, OnTarget Group, Solid Tumors Program, CIMA-CCUN

Tissue context defines cell organization

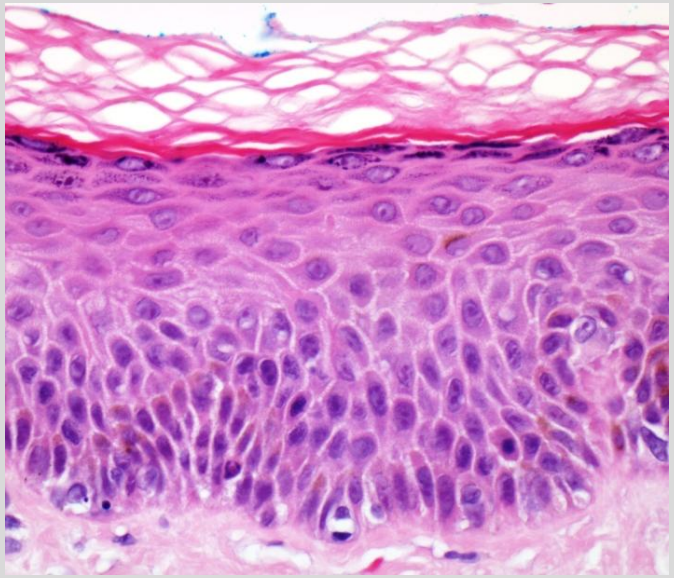
Kidney



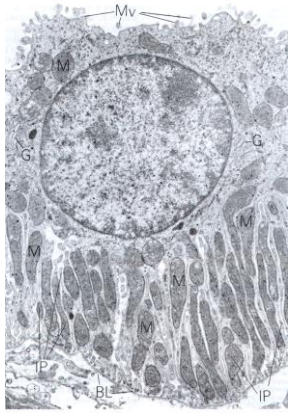
Intestine

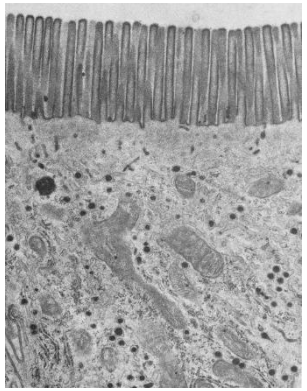


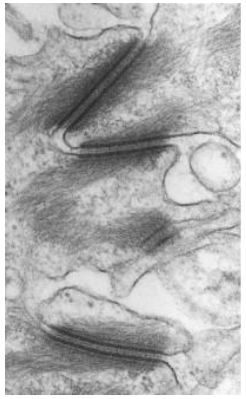
Skin



Membrane specializations according to their function







MEMBRANE SPECIALIZATIONS

Regions of the plasma membrane with specialized structures → defined cell functions (e.g., adhesion, sealing and communication)

CLASSIFICATION

Specializations of the apical membr.

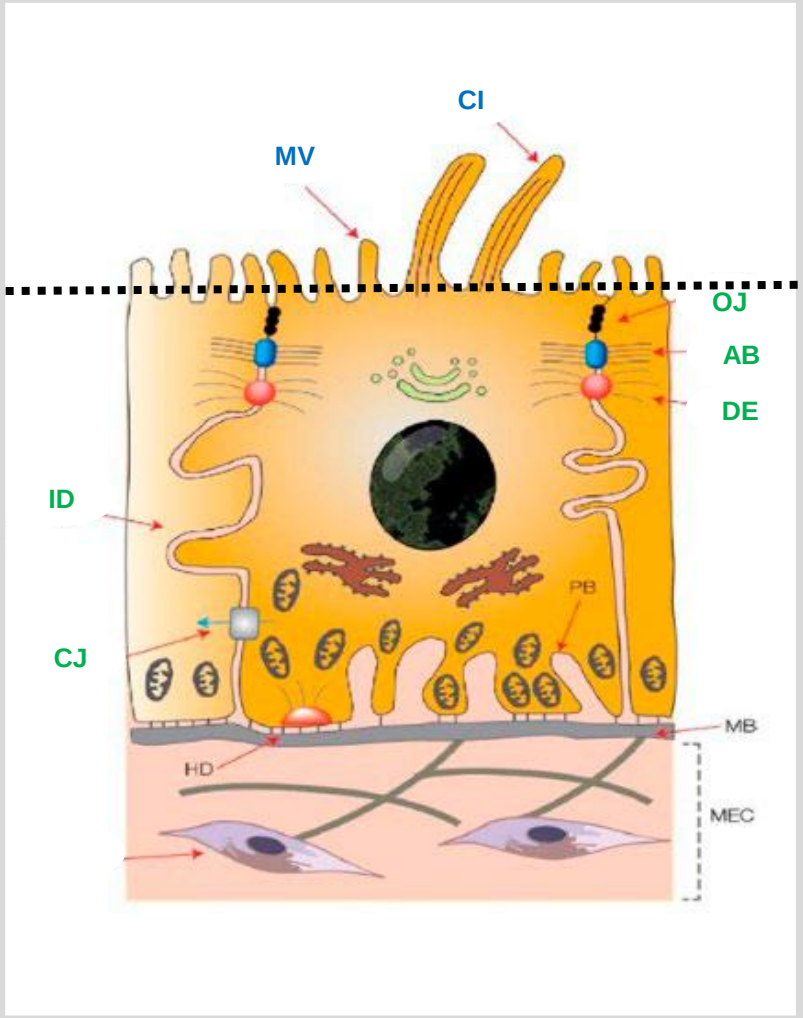
1. Microvilli (MV)

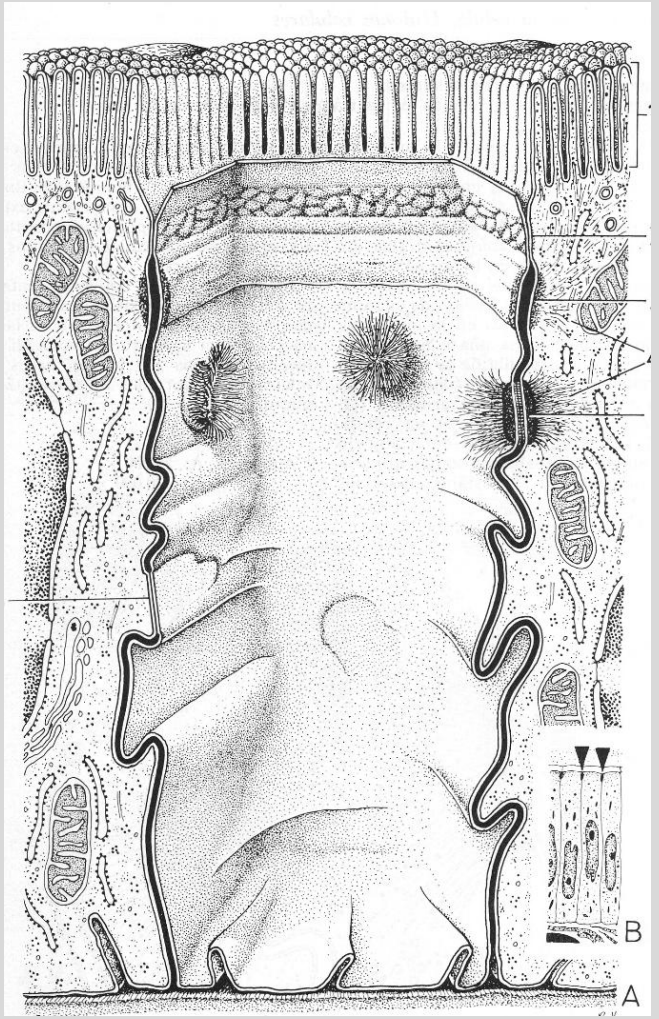
Cilia

Flagella

Specializations of the basolateral membr.

2. Interdigitations/Infoldings (ID)
3. Cell junctions
   - Anchoring junctions
     - Adhesion belt (AD)
     - Desmosomes (DS)
   - Occluding/tight junctions (OJ)
   - Communicating junctions (CJ)





1. MICROVILLI

CHARACTERISTICS

Long projections (apical surface)

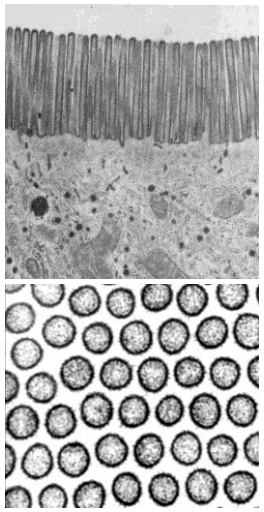
1 µm long and 0.1 µm wide

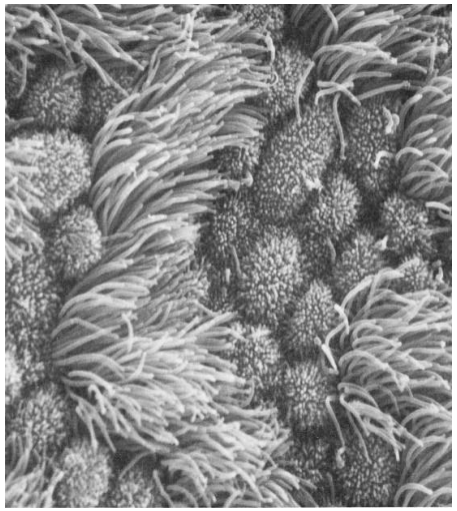
Epithelial cells (e.g., intestine, exocrine glands, respiratory tract)

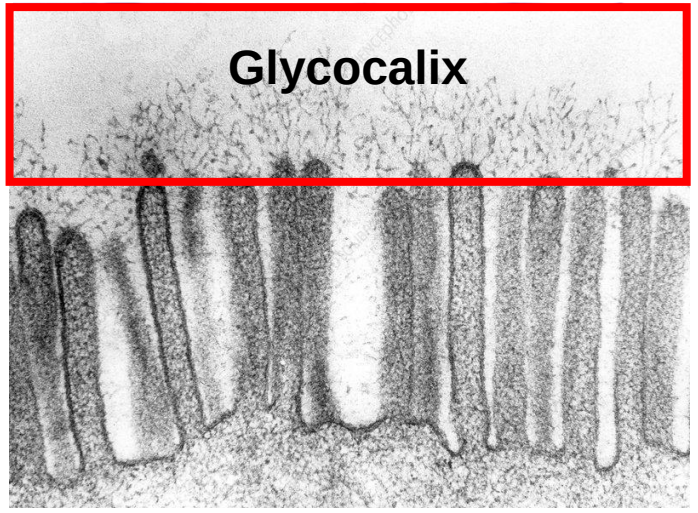
FUNCTIONS

Increase surface area of absorption (i.e., transporter glycoproteins)

Location of digestive enzymes in the glycocalix (protein or sugar digestion)







1. MICROVILLI

ULTRASTRUCTURE

Actin filaments (Actin-F)

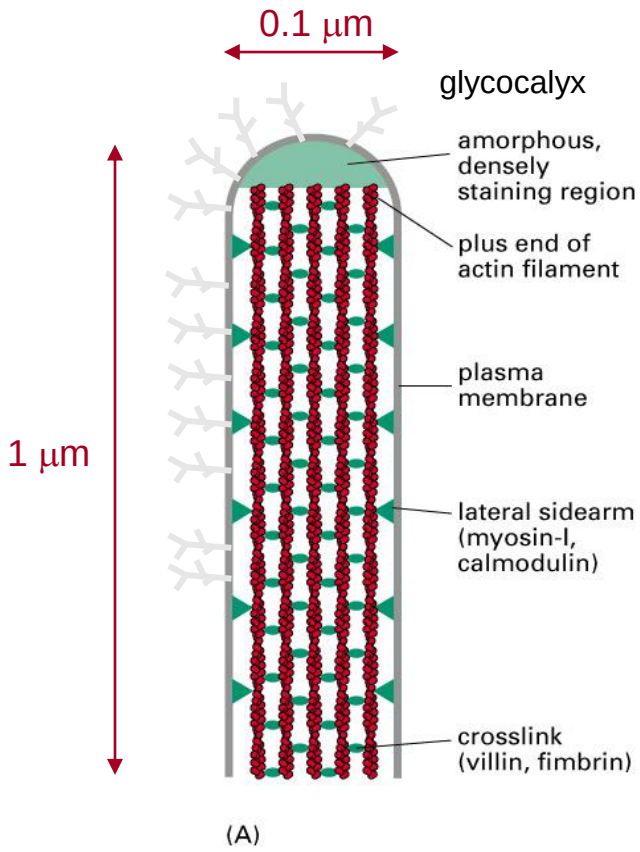
- Longitudinal axis
- 20-30 fibers/bundles

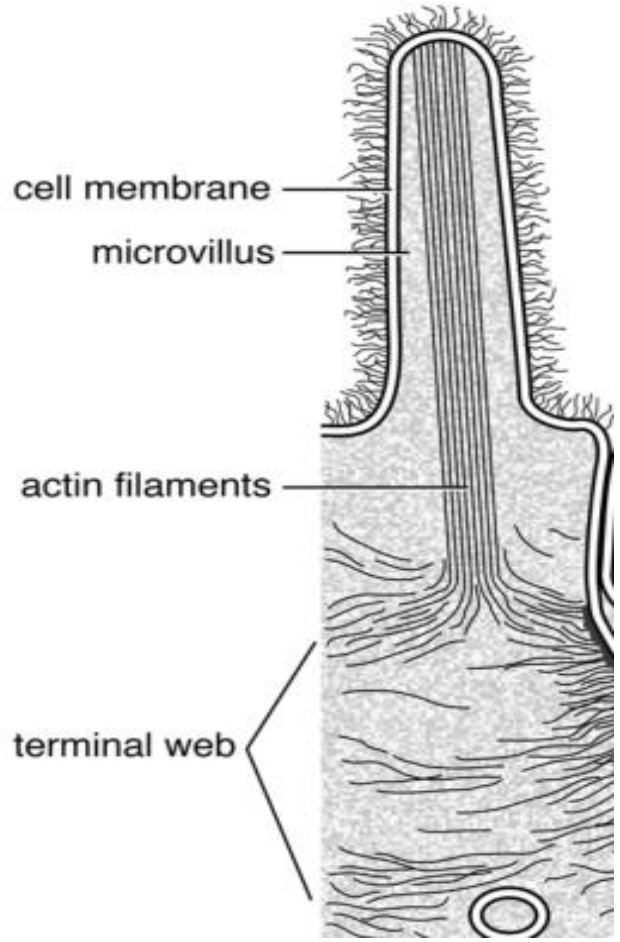
Associated proteins

- Actin-bound (filament cross-link): villin and fimbrin
- Membrane-bound: myosin-I and calmodulin

Terminal web (or terminal veil)

- Transversal axis (parallel to membrane)
- Anchorage of microvilli actin filaments: root foundation
- Myosin, spectrin and intermediate filaments (keratin)





1. MICROVILLI

PATHOLOGICAL CONDITIONS

LACTOSE INTOLERANCE

Reduced lactase concentration →
Lactose is not broken down to glucose and galactose

Consequences: diarrhea, gas, bloating, belly pains and nausea
Treatment: diet changes

HEMOCHROMATOSIS

Increase expression of DMT1 transporter →
Excessive iron build-up in tissues (iron accumulation in heart, endocrine glands - hypophysis, pancreas-, liver, joints)

Consequences: cardiopathy, endocrine/metabolic disorder, arthritis
Treatment: Phlebotomy, diet changes and medication (chelating agents)

2. INTERDIGITATIONS (INFOLDINGS)

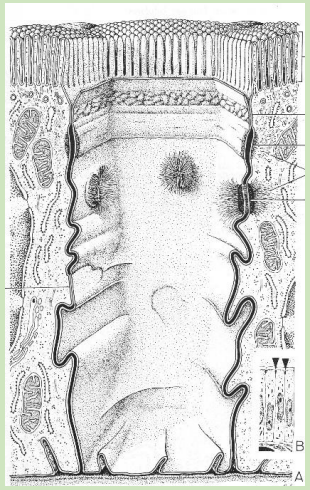
Specializations to increase membrane surface with no increment in cell volumen

LATERAL INFOLDINGS

Characteristics
- Neighboring cells
- Low number

Function
To increase contact surface:
- Stabilize cell from mechanical forces
- Membrane reservoir

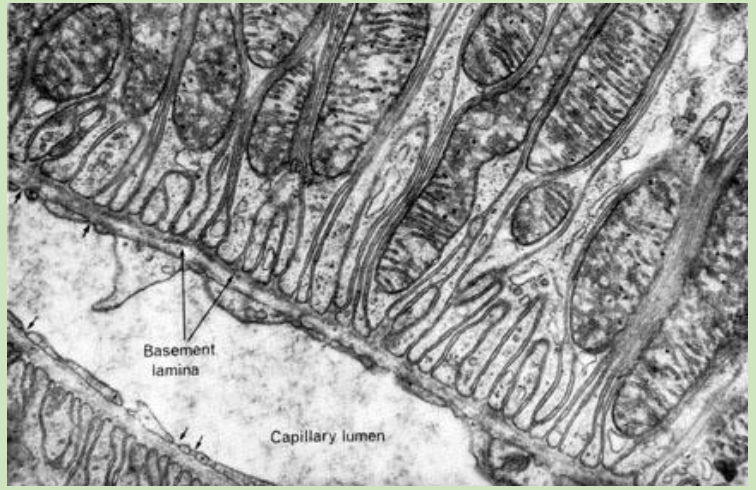




BASAL INFOLDINGS

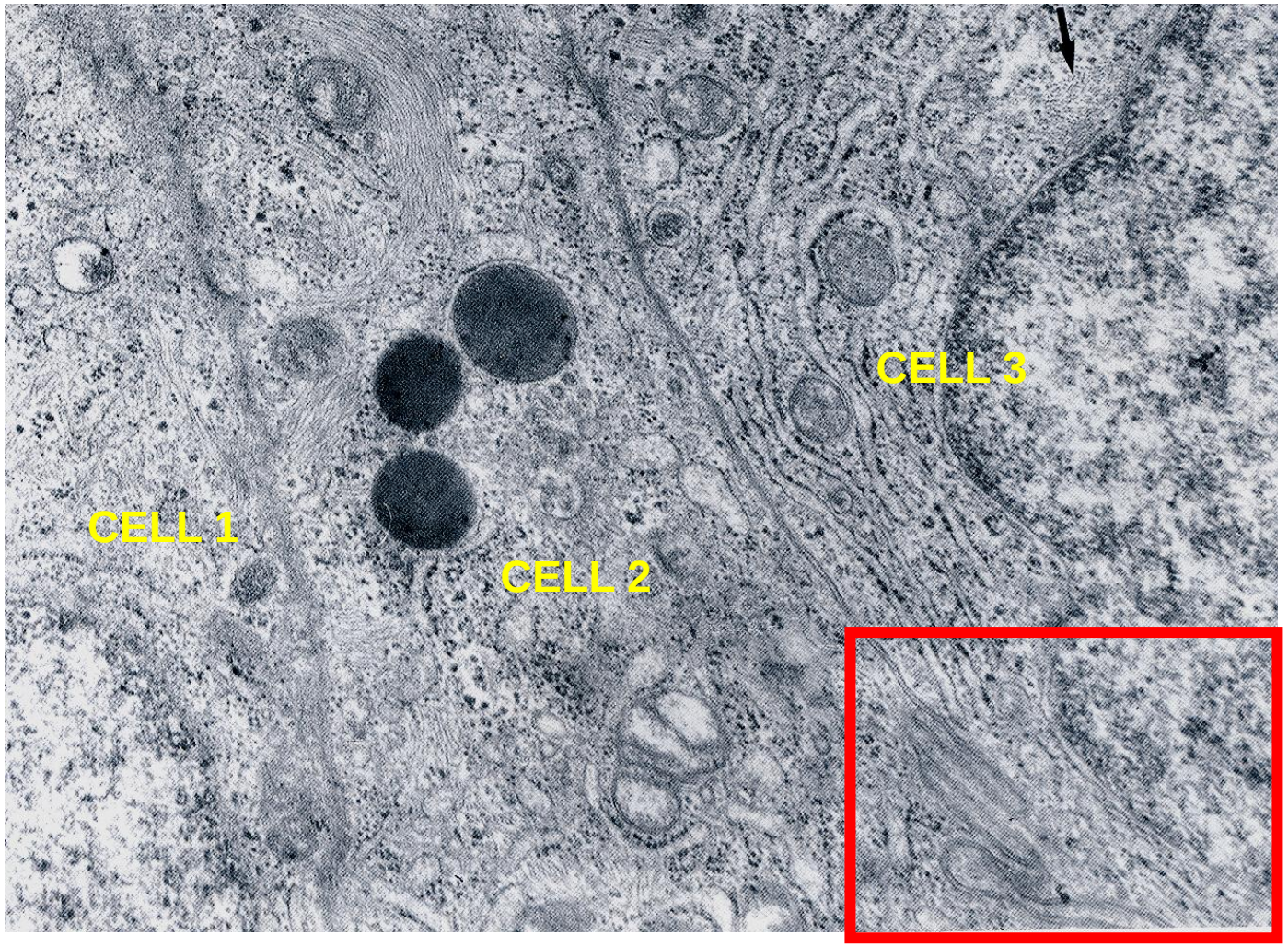
Characteristics
- Cell-to-basal membrane
- Highly developed in certain cell types
- Long projections towards cytoplasm

Function
To increase active ion transport (ATP pumps)
- Epithelial cells with a role in fluid transport (e.g. distal convoluted tubules in kidney)





2. INTERDIGITATIONS (INFOLDINGS)



3. CELL JUNCTIONS

LATIN TERMINOLOGY

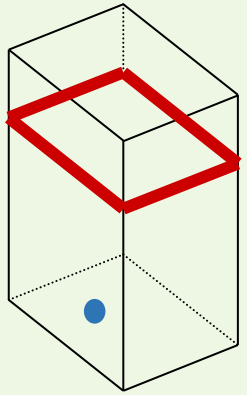
Morphology

Zonula (band)

- zonula adherens (adhesion band)
- zonula occludens (occluding junctions)

Macula (disc-like)

- macula adherens (desmosome)
- macula communicans (communicating junctions)

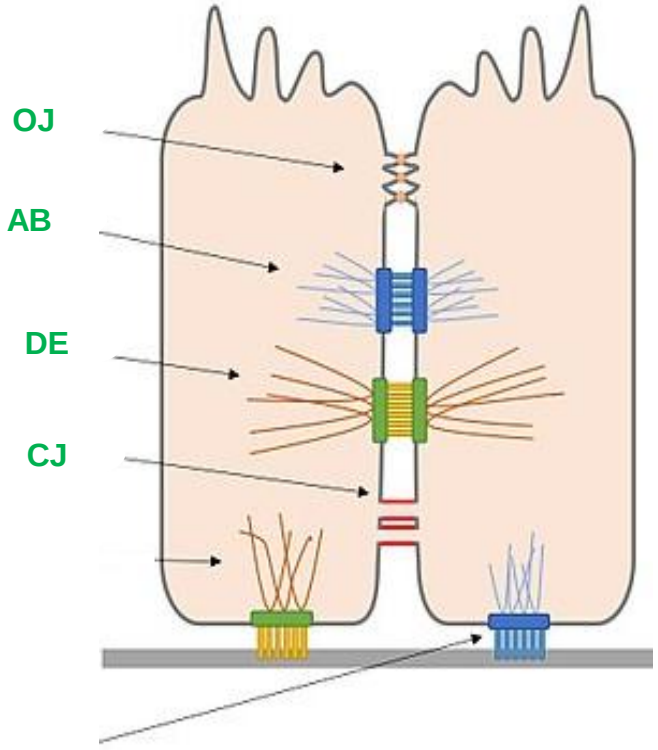


Function

Occludens (sealing, no intercelular space)

Adherens (attachment; separation 20-25 nm)

Communicans (communication)



3.1 ADHERENT JUNCTIONS

GENERAL FUNCTION

Mechanical stability linking the cytoskeleton of one cell to the cytoskeleton of an adjacent cell or to the basal membrane

TYPES OF ADHERENT JUNCTIONS

a. Band-like

Adhesion belt (cell-to-cell)

b. Disc-like

- Desmosomes (cell-to-cell)
- Hemidesmosomes (cell-to-ECM)
- Focal adhesions/contacts (cell-to-ECM)

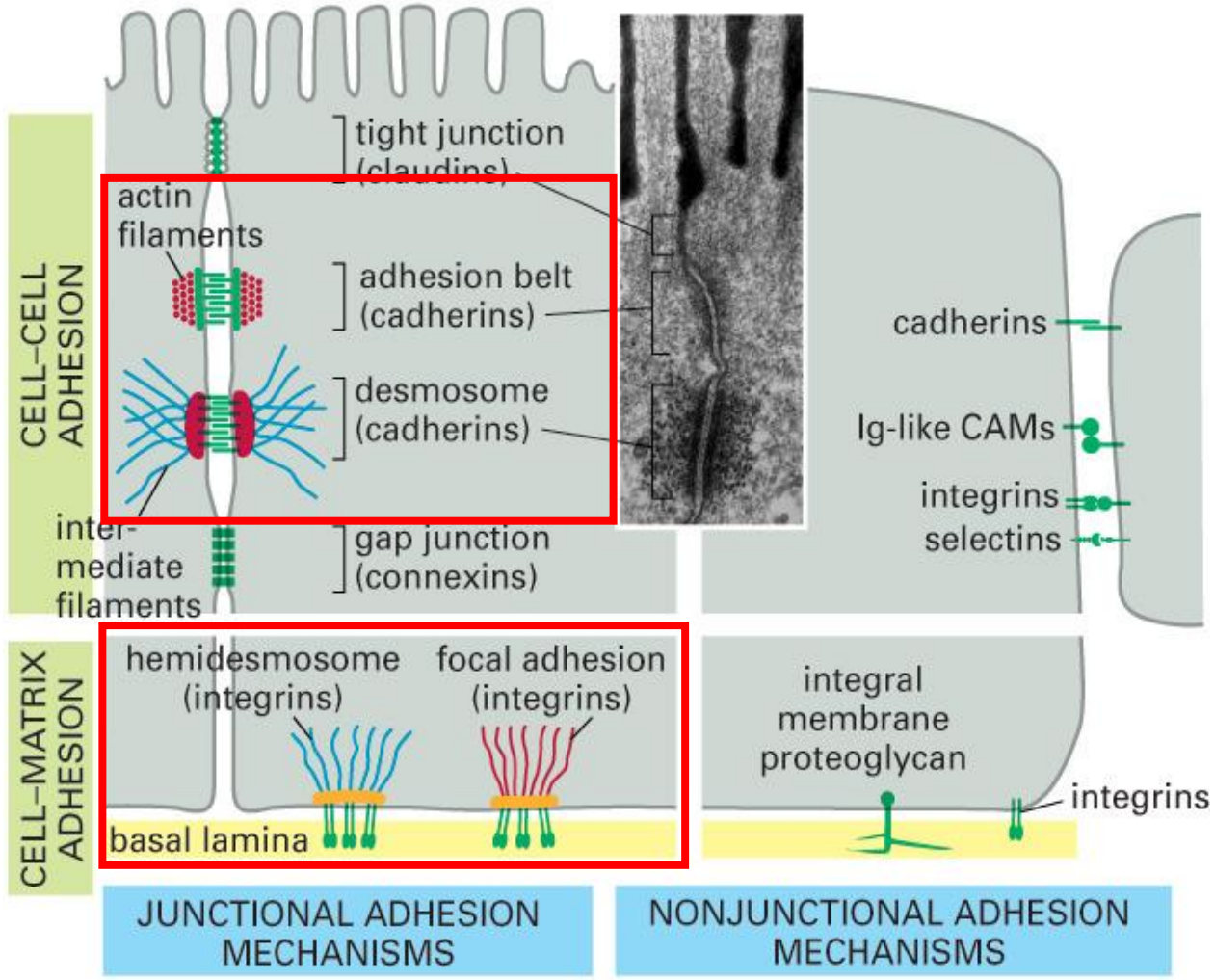


Figure 19–32. Molecular Biology of the Cell, 4th Edition.

3.1.A ADHESION BELT

Adhesion band (zonula adherens)

1 µm width

Towards apical region of epithelial cells (merges with terminal web)

Proteins

- Transmembrane prots (cadherins -homodimers-)

  Classical E-cadherin family

- Intracellular connecting prots

  α and β catenin

- Cytoskeleton prots

  Actin filaments

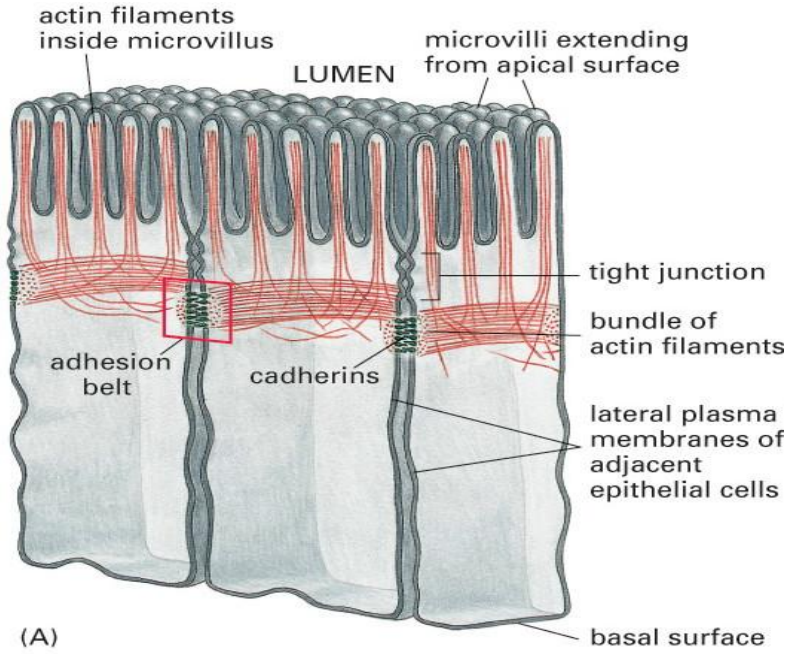
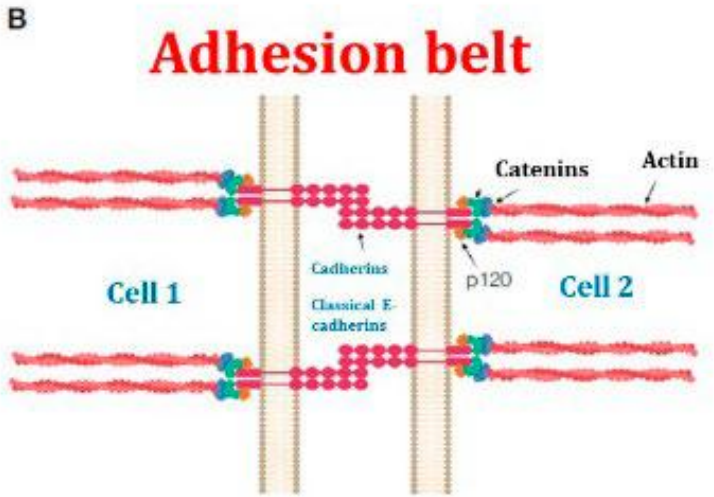


Figure 19–9 part 1 of 2. Molecular Biology of the Cell, 4th Edition.



3.1.B DESMOSOME

Adhesion disc (macula adherens)

1 µm in diameter

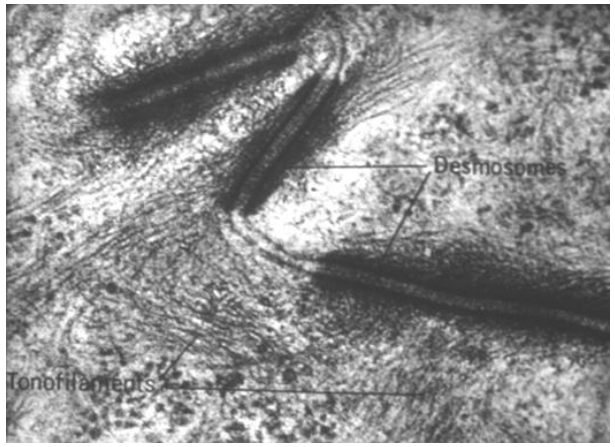
More # desmosomes → Increased mechanical strength

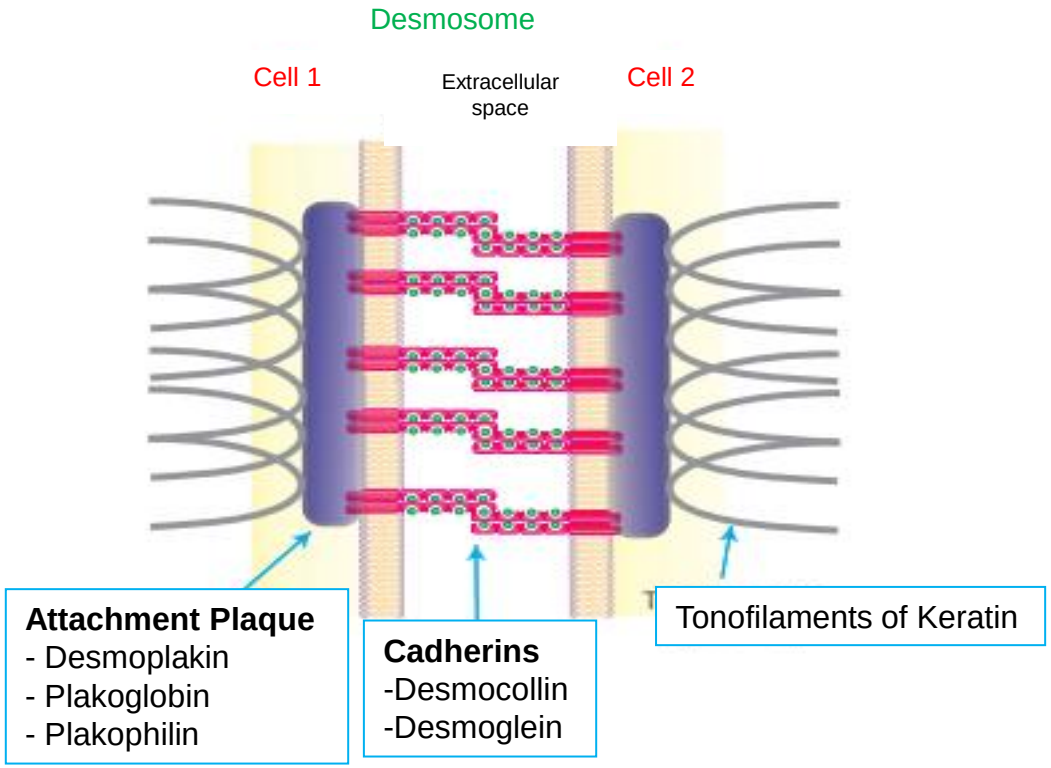
Proteins

- Transmembrane prots (cadherins)
  - Desmocollin
  - Desmoglein

- Intracellular connecting prots
  - Desmoplakin, plakoglobin, plakophilin (attachment plaque)

- Cytoskeleton prots
  - Keratin or desmin (tonofilaments)



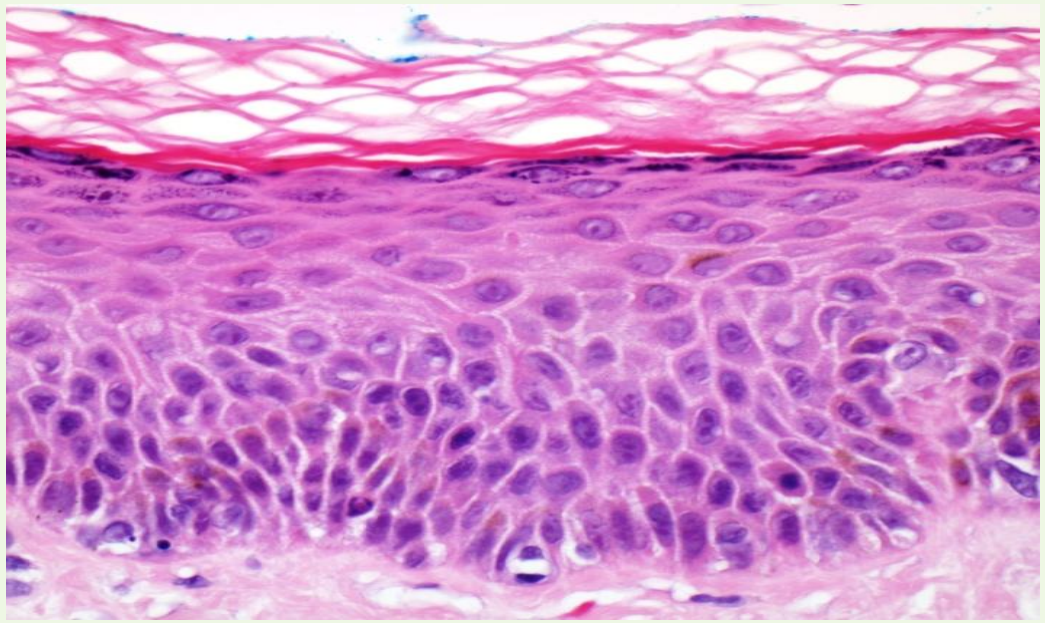


3.1.B DESMOSOME

Functional role: Mechanical resistance

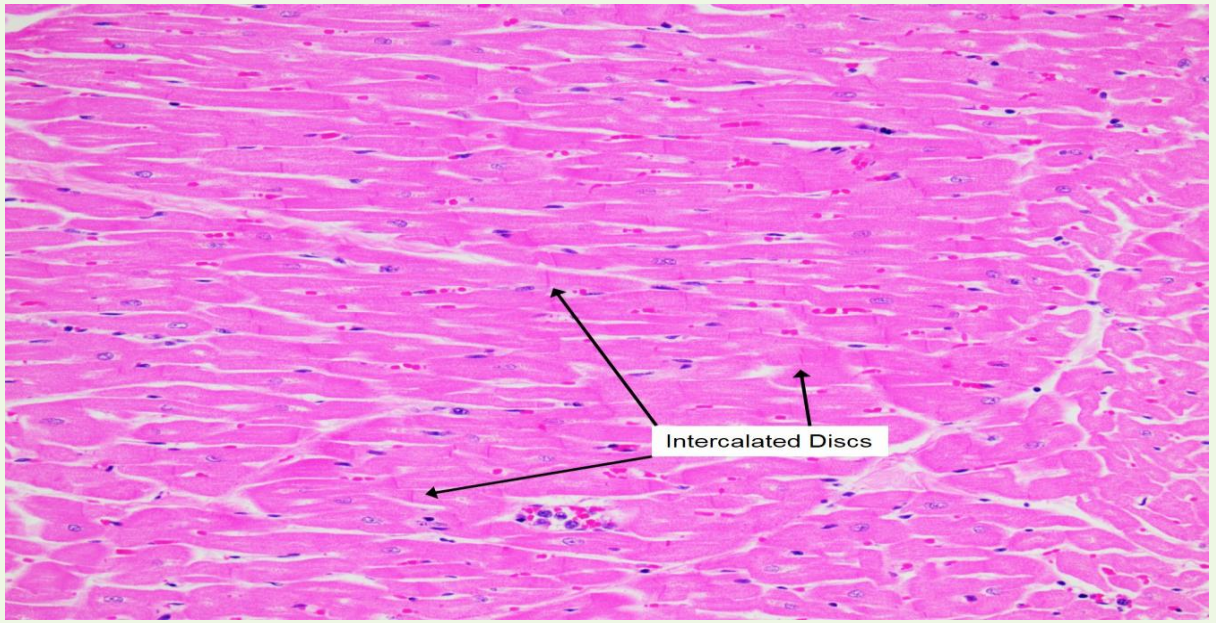
EPIDERMIS

Spinous cells
Cytoskeletal protein: keratin



CARDIAC MUSCLE

Cardiomyocytes
Cytoskeletal protein: desmin



3.1.B DESMOSOME

PATHOLOGIES RELATED TO DESMOSOME ALTERATIONS

PEMPHIGUS VULGARIS

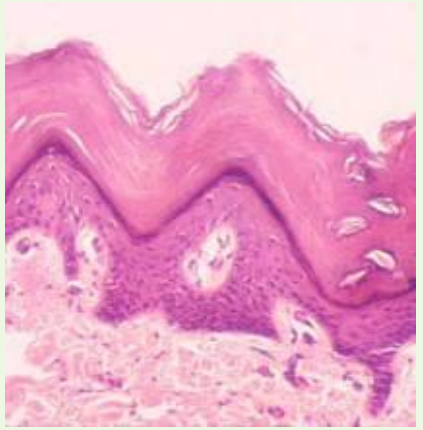


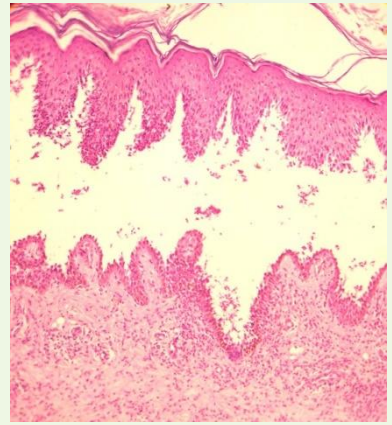
- Autoimmune disease:

Antibodies against desmoglein 1, 3 or both

Spinous cells

- Acantholysis: separation of skin layer→ blisters
- Skin is prone to infections





3.1.C HEMIDESMOSOME

Adhesion disc

1 µm in diameter

½ of a desmosome

Proteins

- Transmembrane prots

  Integrin heterodimer: α6/β4

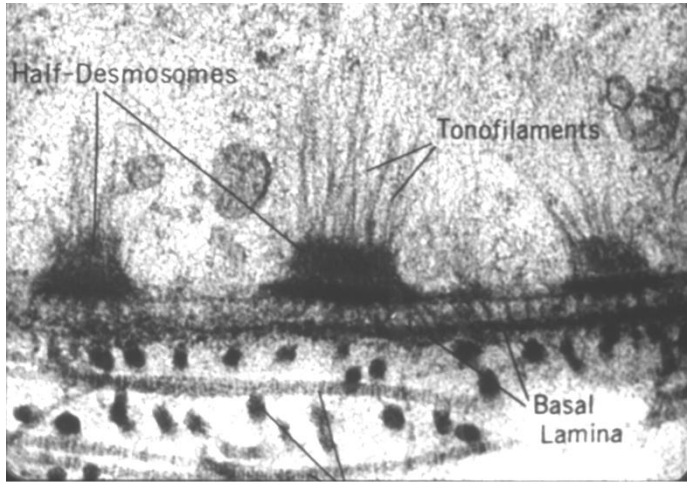
- Intracellular connecting prots

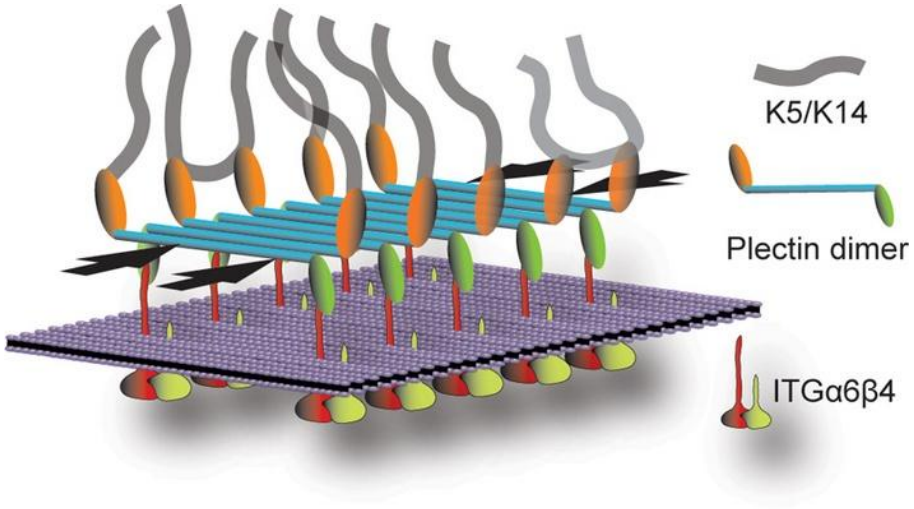
  Plectin

  BP230

- Cytoskeleton prots

  Keratin 5 and 14





3.1.C HEMIDESMOSOME

PATHOLOGIES RELATED TO DESMOSOME ALTERATIONS

BULLOUS PEMPHIGOID

Autoimmune response to BP230 protein
No acantholysis



FIGURE 1: Clinical Lesions

Diffuse and erythematous tense bullae

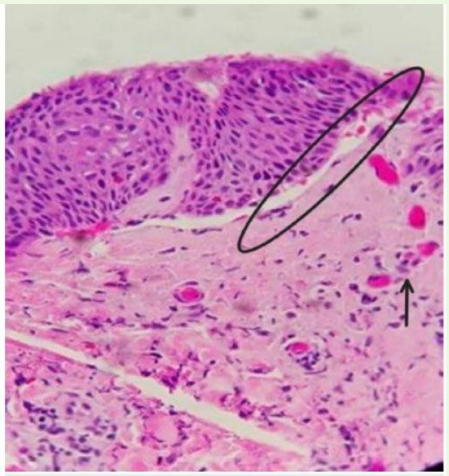
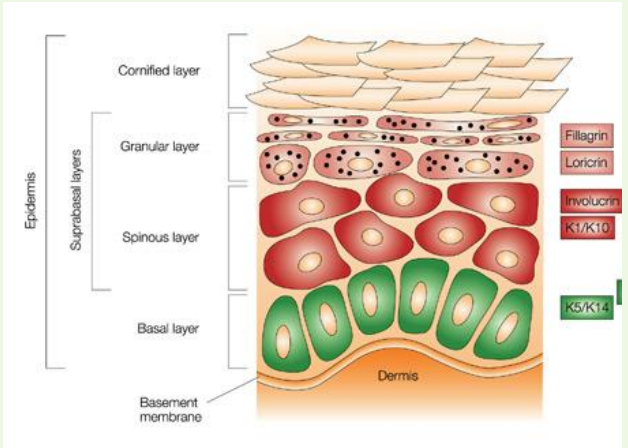


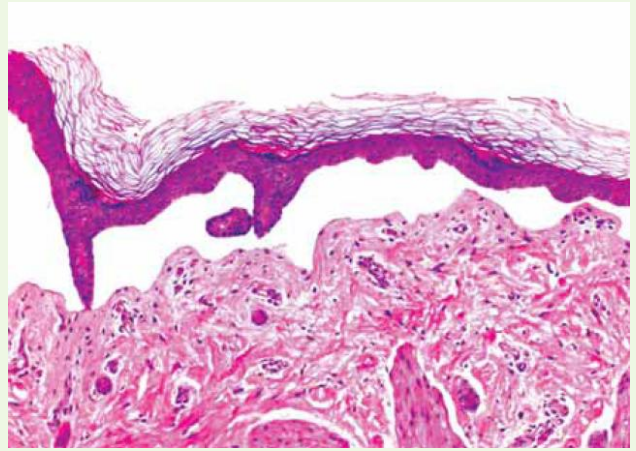
FIGURE 2: Histology

Subepidermal blister (oval) with eosinophils and neutrophils and underlying dermis demonstrating severe edema and infiltrate composed of eosinophils and lymphocytes (arrow)

EPIDERMOLYSIS BULLOSA

Mutations in K5 and K14 genes
Presents with acantholysis (basal layer cells)





3.1.D FOCAL CONTACT

Adhesion disc (macula adherens)
0.5 -1 uM in diameter

Proteins

Transmembrane prots

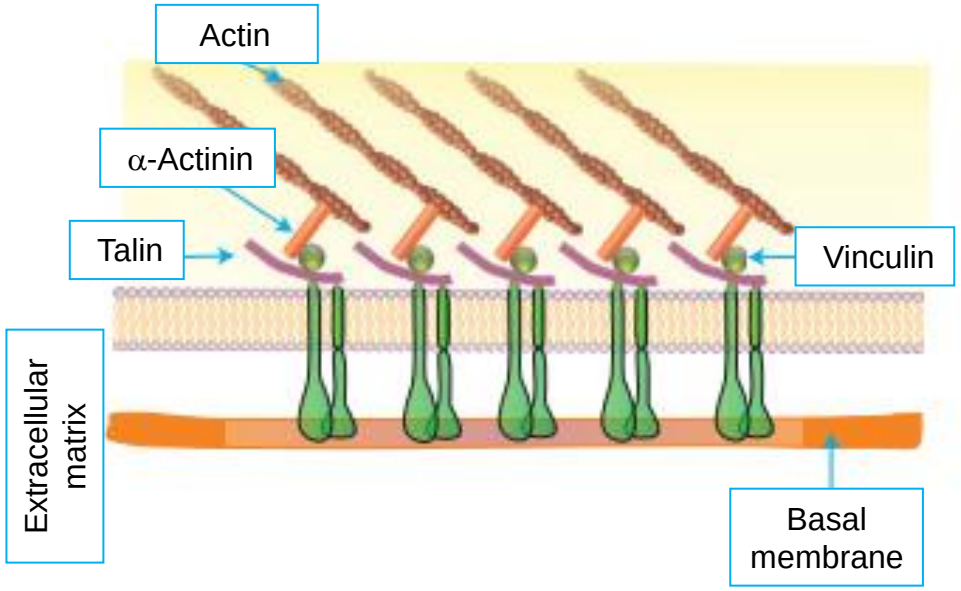
Integrins

Intracellular connecting prots

Talin and **vinculin**

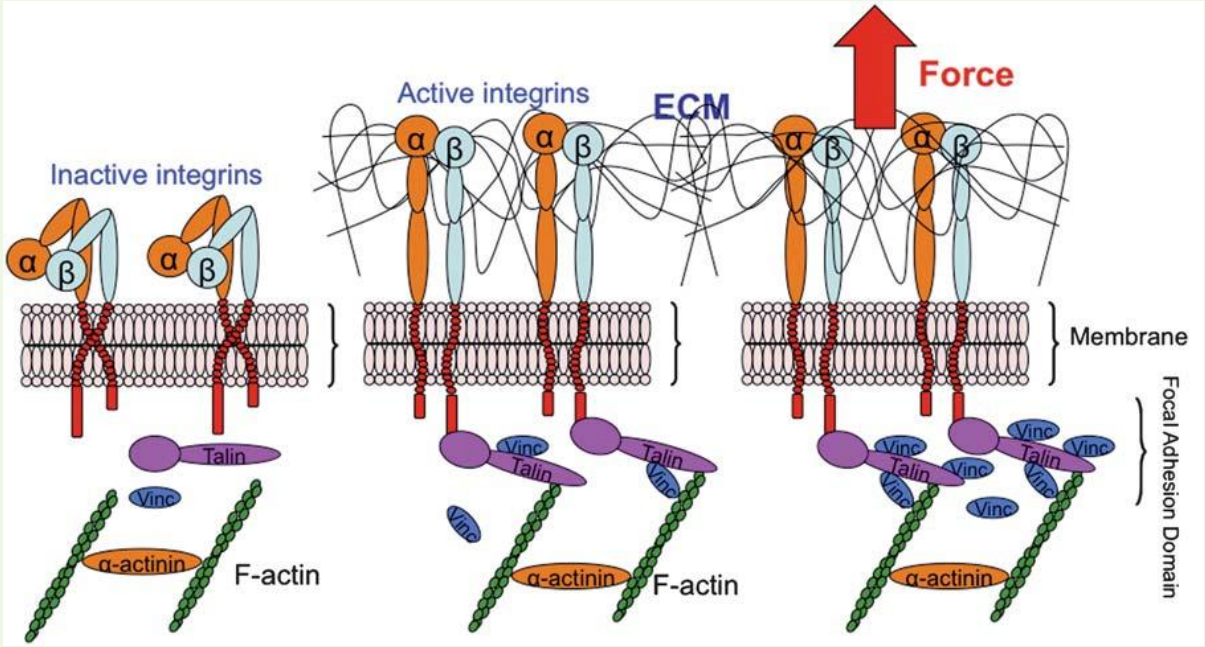
Cytoskeleton prots

Actin



Functional role

- Cell motility

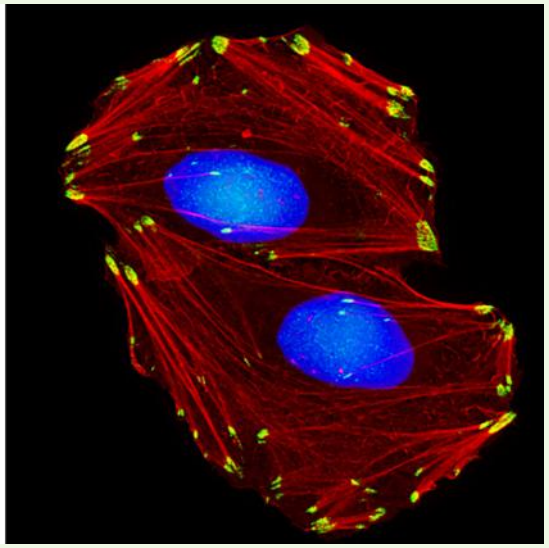


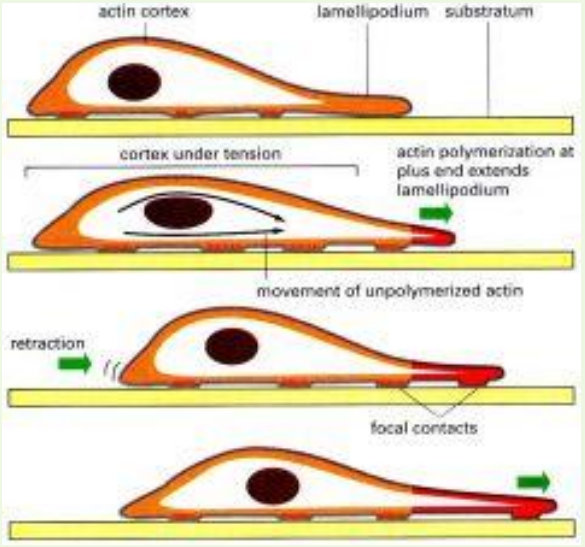
Nuclei - blue

Focal Adhesions - Green

Actin - red

Actin fibers are anchored to the cell membrane at focal contacts

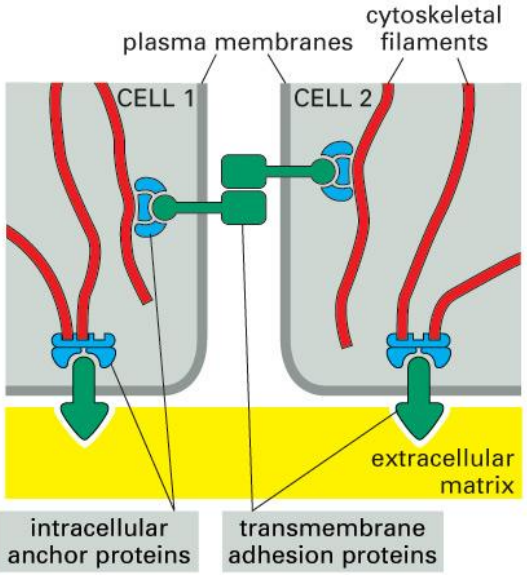


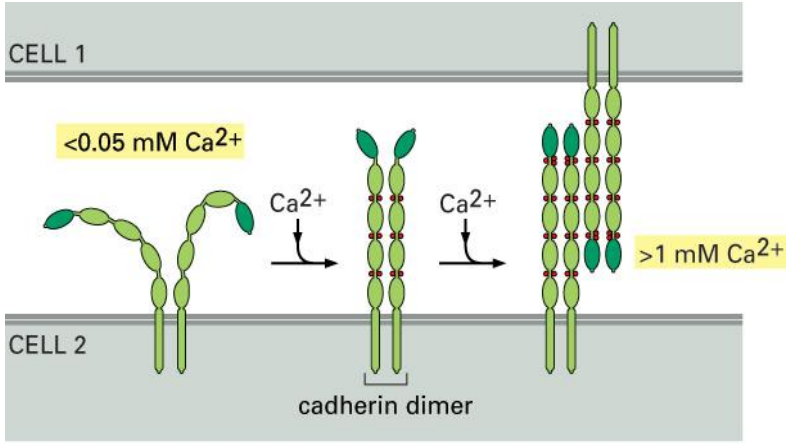


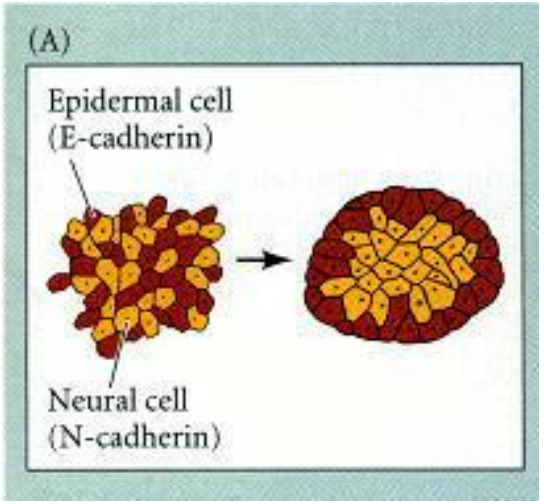
TYPES OF TRANSMEMBRANE ADHERENT PROTEINS

CADHERINS

- Cell-to-cell binding
- Homodimers (homophilic interaction)
- Ca^{2+} dependent
- Tissue specific
- Specific to adherent junctions
  - Adhesion band: Classical E-cadherin
  - Desmosomes: Desmoglein and desmocollin







TYPES OF TRANSMEMBRANE ADHERENT PROTEINS

INTEGRINS

- Cell-to-ECM binding (cell receptors to ECM prots)

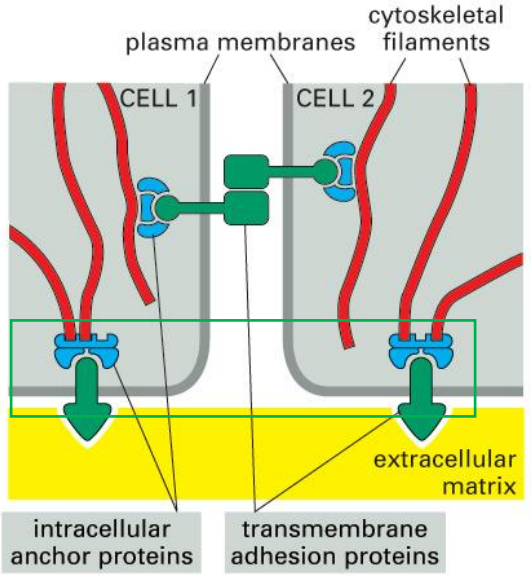
- Heterodimers (α and β subunits)

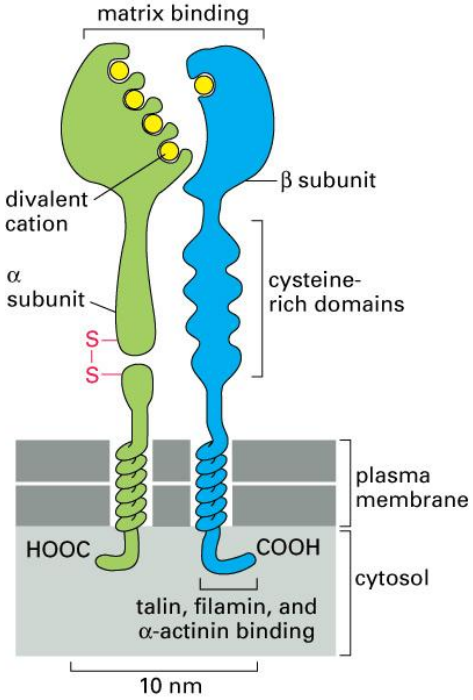
- ECM prots: laminin (basal mbr.) and fibronectin (heterophilic interactions)

- Ca^{2+} dependent

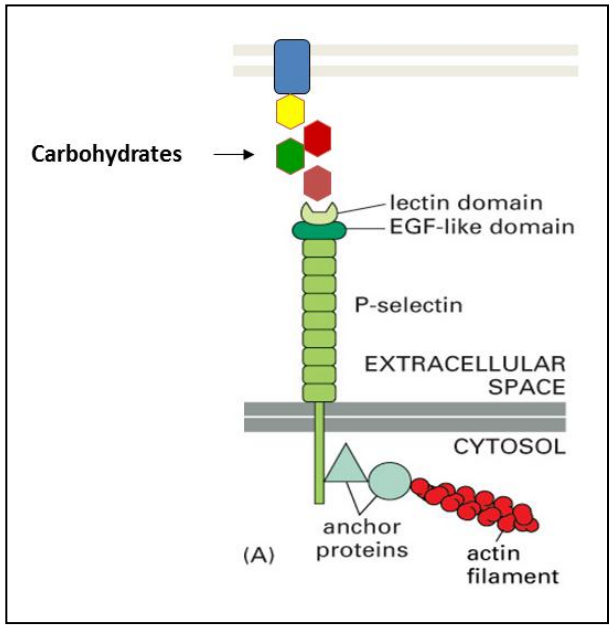
- Important for intracellular signaling (e.g., focal adhesion kinases FAK)

- Transient or permanent adhesion structure





ADDITIONAL TRANSMEMBRANE PROTEINS FOR TRANSIENT CELL ADHESION



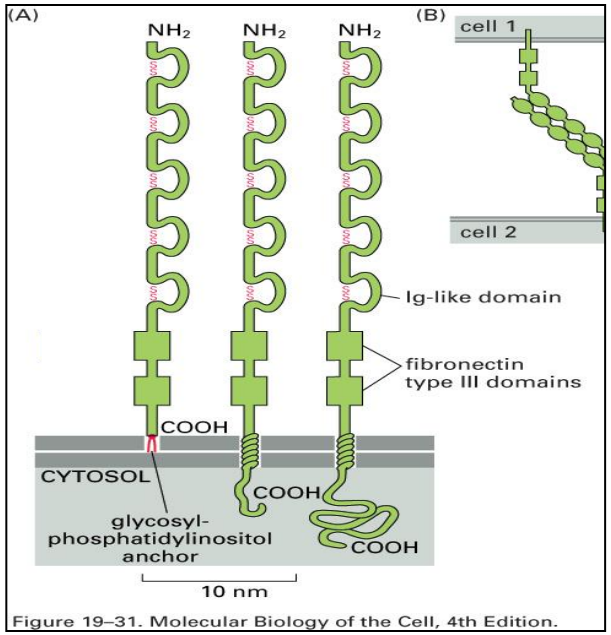


Figure 19–31. Molecular Biology of the Cell, 4th Edition.

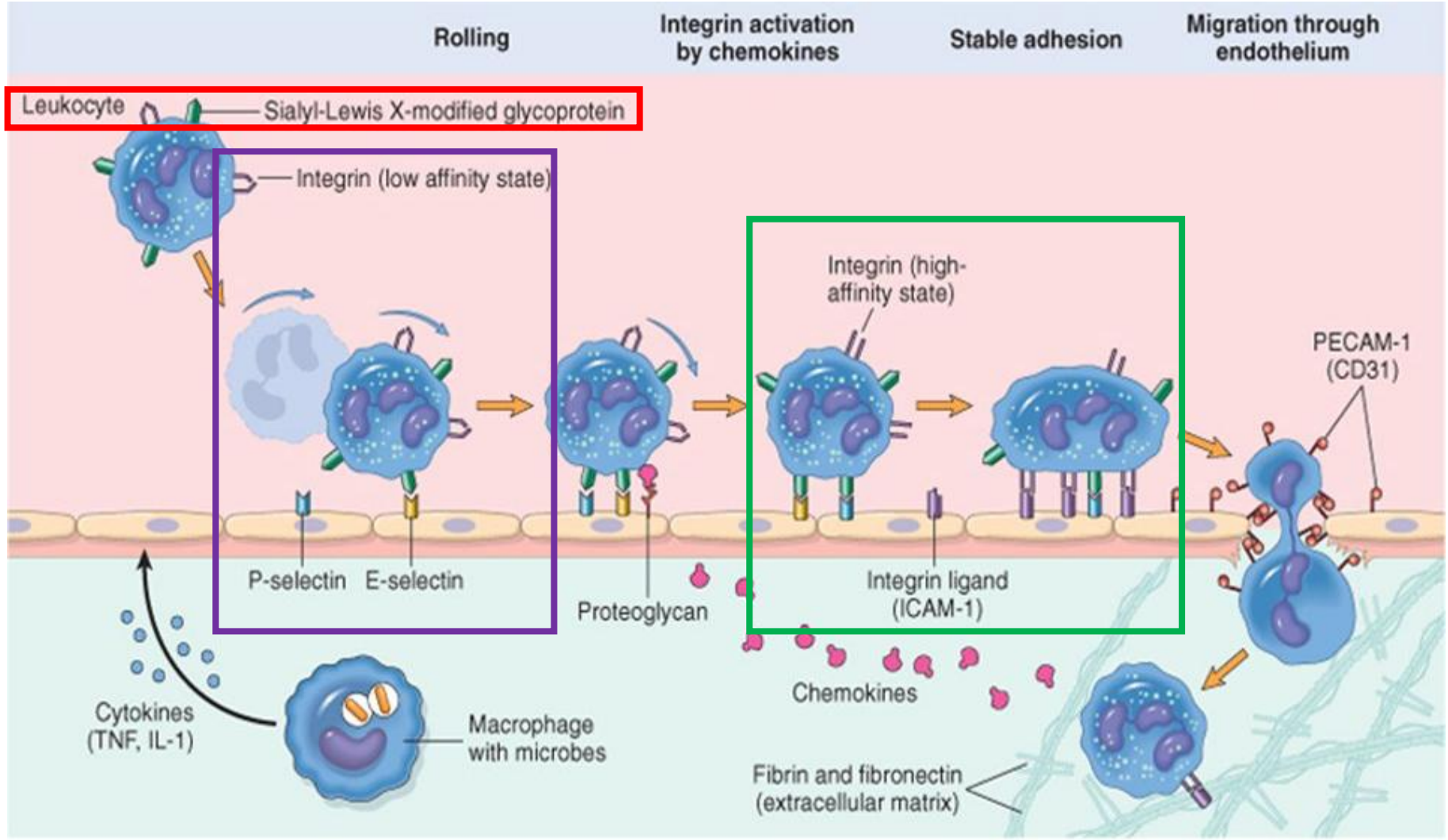
Selectins

- Monomers that form heterophilic interactions with **sugar chains** in glycoproteins
- Ca^{+2}-dependent
- Important for leukocyte-endothelial interactions

IgSF-CAM or ICAM

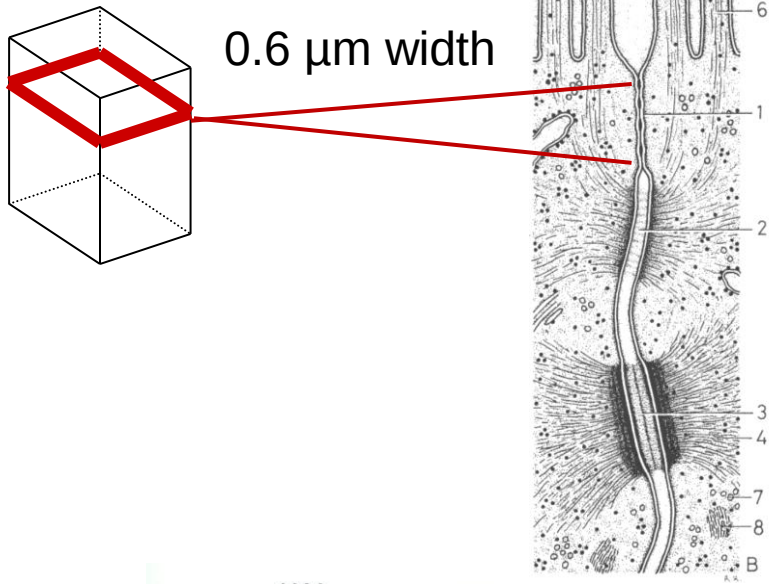
- **I**mmunoglobulin **S**uper **F**amily **C**ell **A**dhesion **M**olecules
- Monomers that form homophilic (other **IgSF-CAM**) or heterophilic interactions (**integrins**)
- Ca^{+2}-**independent**
- Important for leukocyte-endothelial interactions

TRANSENDOTHELIAL MIGRATION OF LEUKOCYTES



TYPE OF CELL ADHESION MOLECULE	INTERACTION TYPE	HOMOPHILIC-HETEROPHILIC INTERACTION	STABLE-TRANSITORY
CADHERIN HOMODIMER	CELL-TO-CELL	HOMOPHILIC	STABLE (Adhesion belt /desmosomes)
INTEGRIN HETERODIMER (α/β)	CELL-TO-ECM	HETEROPHILIC (ECM)	STABLE (hemidesmosomes/ focal contacts)
SELECTIN MONOMER	CELL-TO-CELL	HETEROPHILIC (IgSF oligosacharides and glycolipids)	TRANSITORY (leucocyte extravasation)
IgSF-CAM (e.g. ICAM)	CELL-TO-CELL	HOMOPHILIC HETEROPHILIC (Integrins)	TRANSITORY (leucocyte extra.)
Transitory INTEGRINS	CELL-TO-CELL	HETEROPHILIC (IgSF-CAM)	TRANSITORY (leucocyte extra.)

3.2 OCCLUDING / TIGHT JUNCTIONS



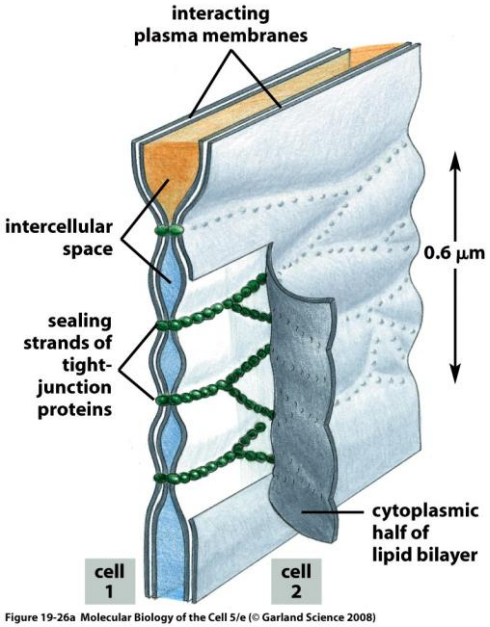
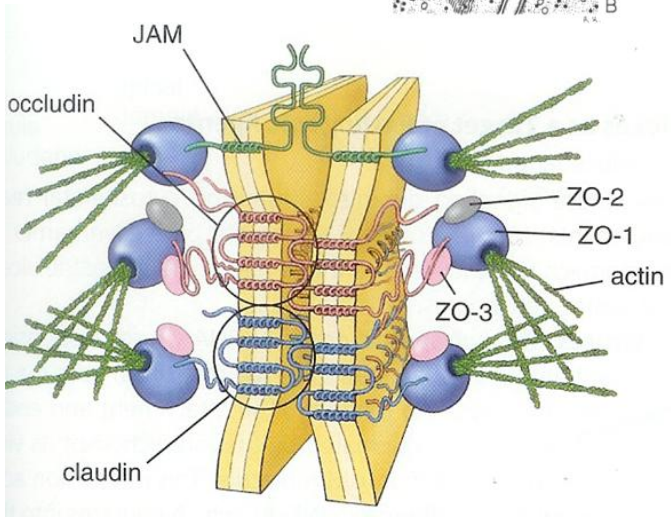


Figure 19-26a Molecular Biology of the Cell 5/e (© Garland Science 2008)



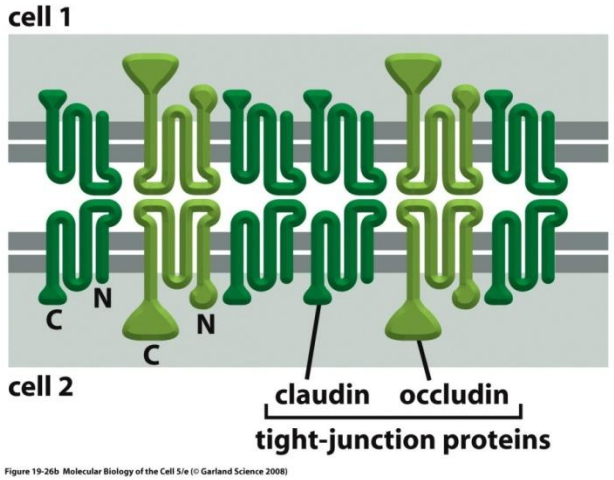


Figure 19-26b Molecular Biology of the Cell 5/e (© Garland Science 2008)

FUNCTION: seal intercellular space

PROTEINS

Transmembrane (homodimers)

- **Claudin**
- **Occludin**
- **JAM** (junction adhesion molecules)
- Nectin

Cytoplasmic

- Zonula Occludens prots (**ZO-1, 2, 3**)
- Afadin

Cytoskeleton proteins

- Actin filaments

JUNCTIONAL COMPLEX

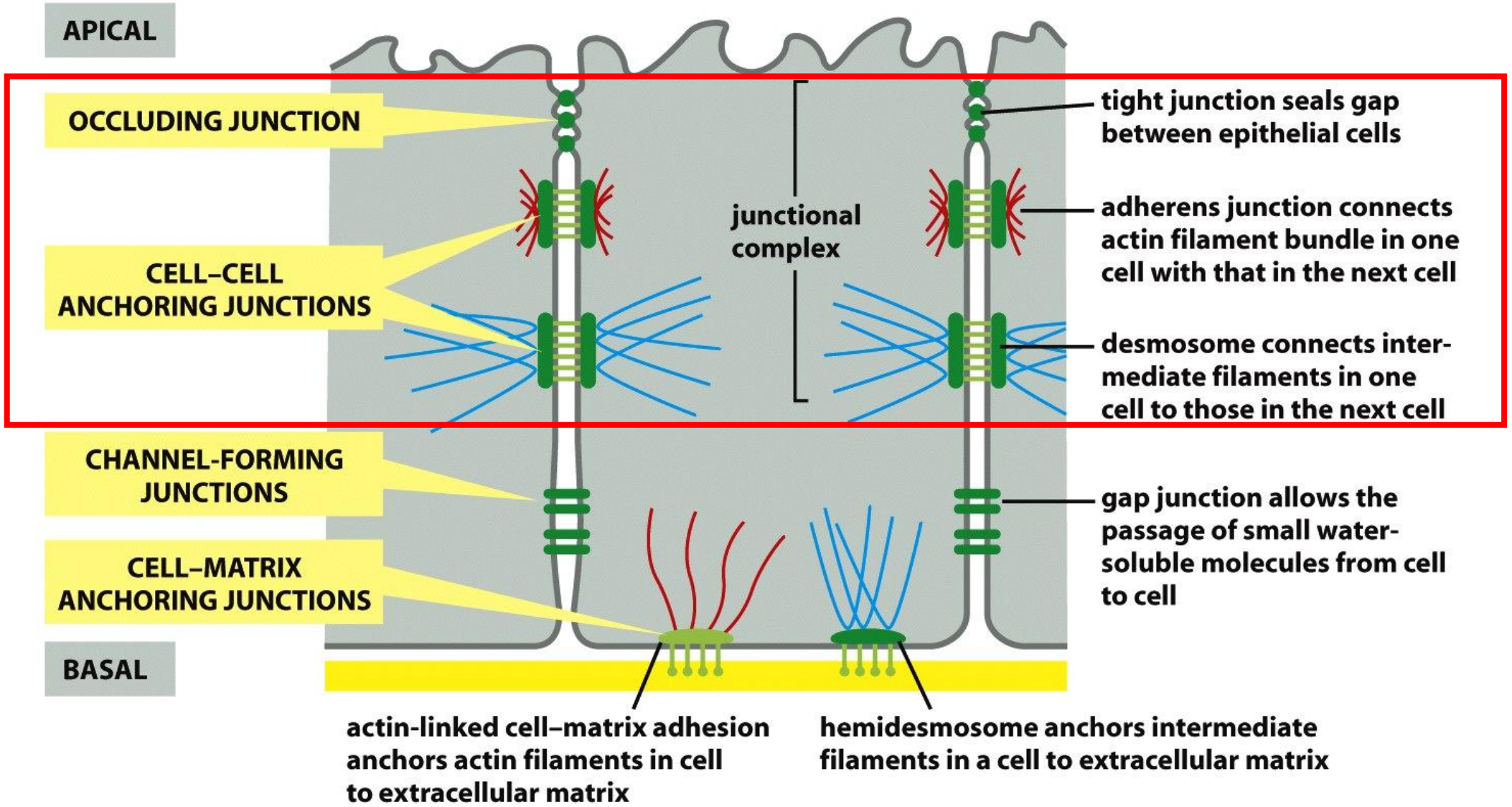
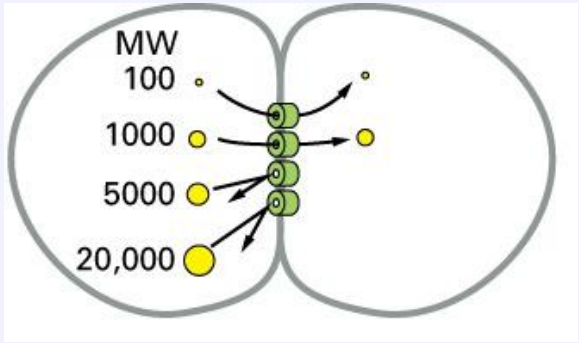


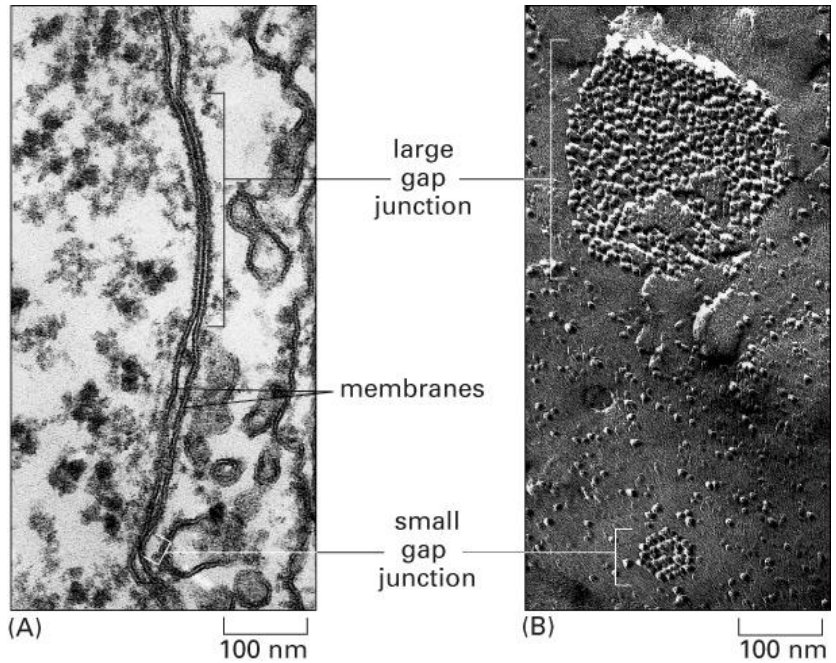
Figure 19-3 Molecular Biology of the Cell 5/e (© Garland Science 2008)

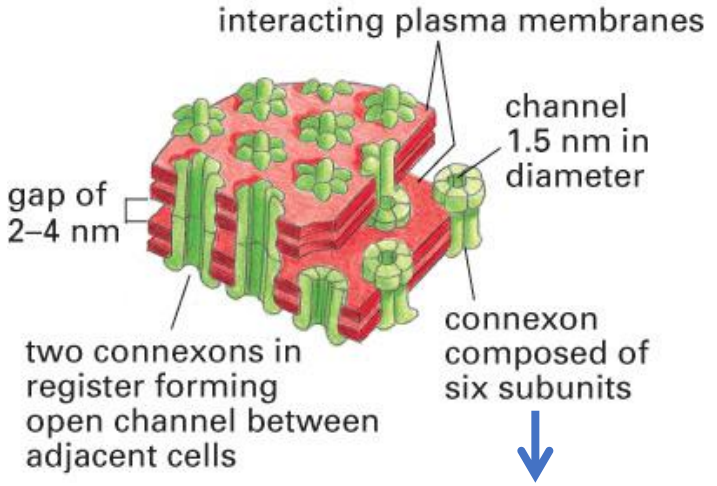
3.3 COMMUNICATING (GAP / NEXUS)JUNCTIONS

- Disc-like (200-300 µm diameter): multiple channels
- Pores called **Connexons** (one from each cell)
- Direct communication between cells
- Small molecules (<u>≤1.2 nM in diameter; 1 kDa of MW</u>)

  ATP, cAMP, Na^+, Ca^{2+}, aa, vitamins
- High Ca^{2+} closes connexons







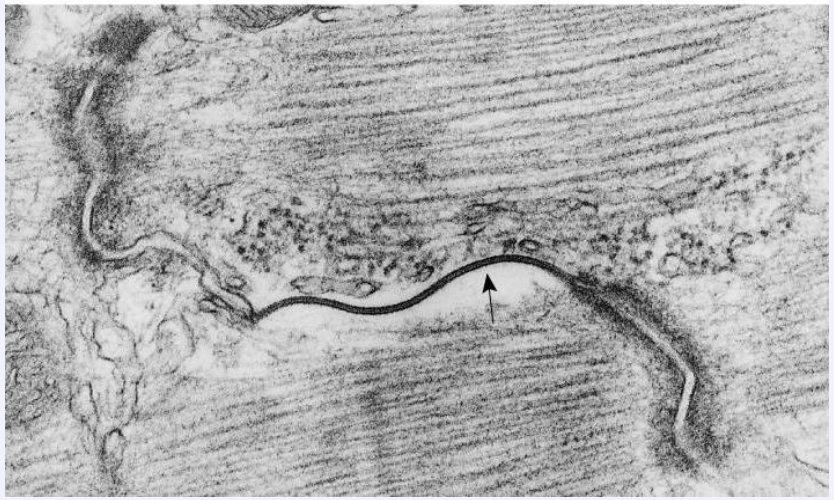
Proteins: **connexins (6)**

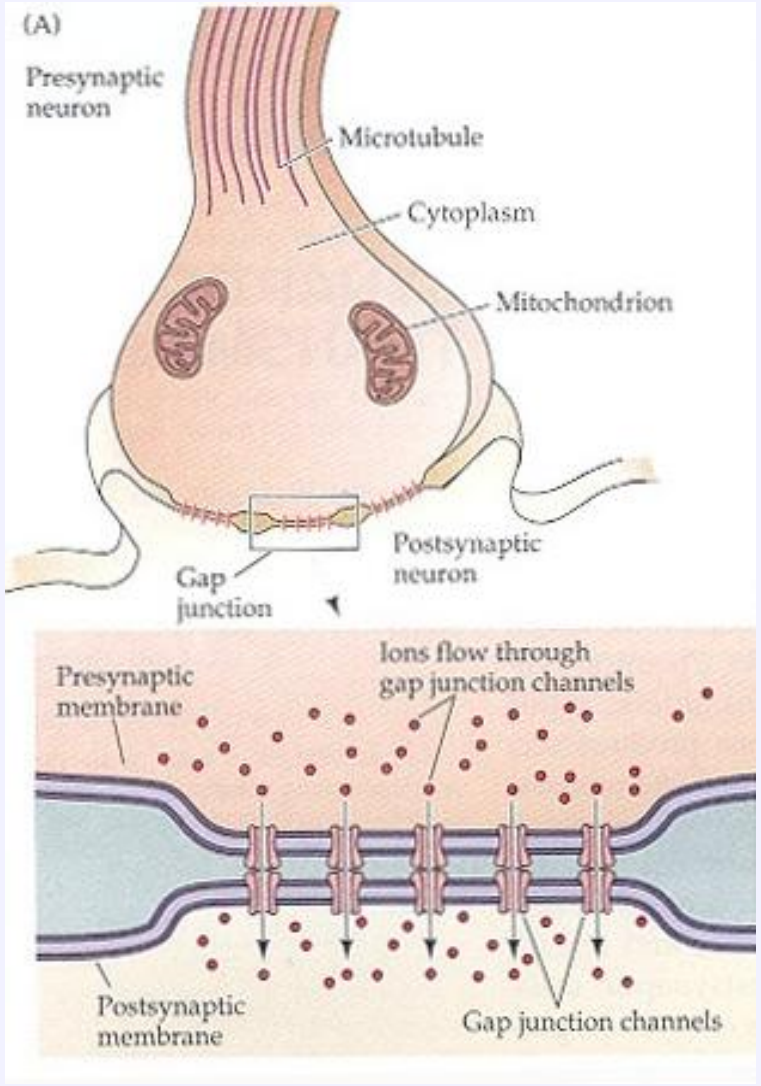
3.3 COMMUNICATING (GAP / NEXUS) JUNCTIONS

Function

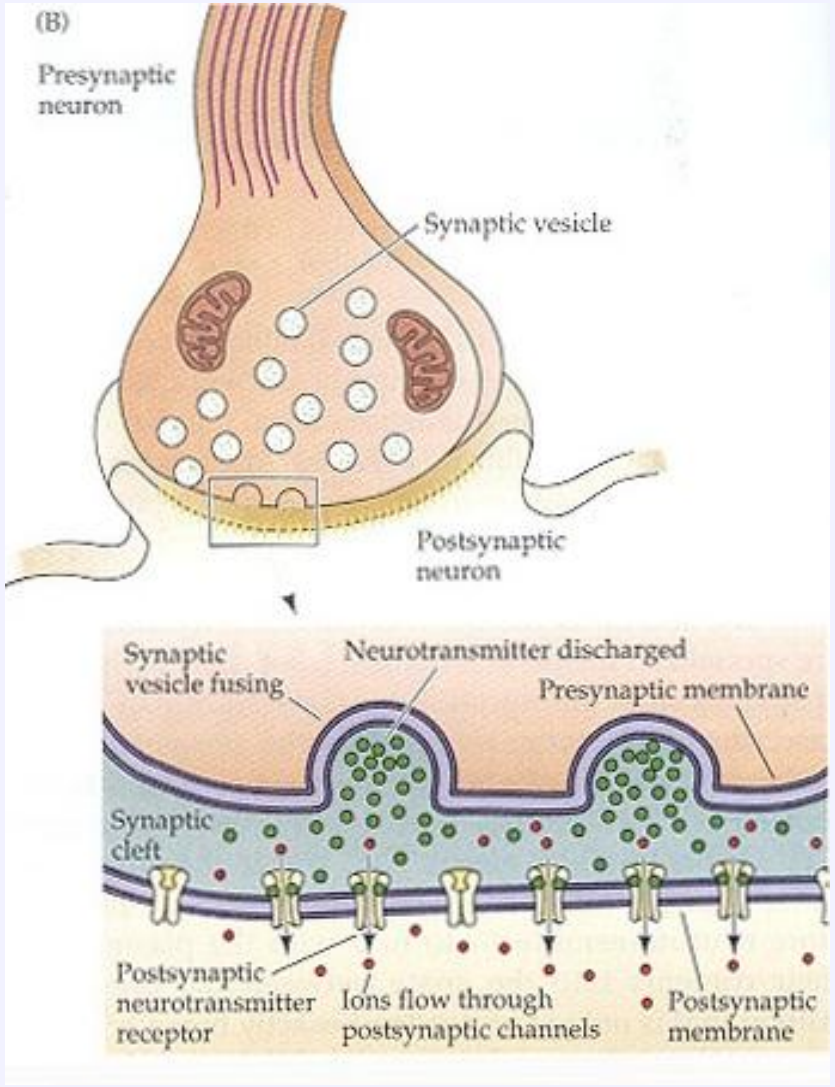
- Cardiomyocytes:
syncronous and
high speed excitation

- Neurons:
Electrical synapses





Electrical synapse



Chemical synapse